Cleaning and EDA

Ray Chandonnet & Barrett Viator

2025-08-29

```r
# REC code

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readxl)
library(writexl)
library(skimr)
library(knitr)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```r
library(ggpmisc)
```

```
## Loading required package: ggpp
## Registered S3 methods overwritten by 'ggpp':
##   method                  from
##   heightDetails.titleGrob ggplot2
##   widthDetails.titleGrob  ggplot2
##
## Attaching package: 'ggpp'
##
## The following object is masked from 'package:ggplot2':
##
##     annotate
```

```r
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```r
library(here)
```

```
## here() starts at /Users/raychandonnet/Chandonnet Family Dropbox/Chandonnet Family Folder/Ray Personal
```

```r
# This module cleans and joins the raw data, saves the cleaned data for use
# by preprocessing, and performs all Exploratory Data Analysis
#
# Read in data
RawData <- readRDS(here::here("Data","PreEDA_DataFrame.rds"))
names(RawData) <- make.unique(names(RawData), sep = ".") # ensure unique column names
FieldActions <- read_excel(here::here("Data", "FieldActions.xlsx"))
CleanedData <- RawData # make a copy to preserve original
IncomeData <- read_excel(here::here("Data", "Income Group Data.xlsx")) # Import income data

# This code iterates through every field in the raw data, taking the cleanup action on
# it that is specified in the FieldActions data frame.  The net result is that a bunch of
# duplicated and/or unneeded fields are deleted, fields with uninterpretable (coded) column
# names tied to their source are renamed to be descriptive, and the Year column is forced
# to an integer.  This was done to automate the cleaning process and make the code more compact.

for (fieldcounter in names(CleanedData)) {
  action_row <- FieldActions[FieldActions$FieldName == fieldcounter, ]
  if (nrow(action_row) == 0) {
    next  # No action specified for this field
  }
  WhatToDo <- action_row$Action # Read the field's action and act accordingly
  if (WhatToDo == "Delete") {
    CleanedData[[fieldcounter]] <- NULL # Drop

  } else if (WhatToDo == "Rename") {
    names(CleanedData)[names(CleanedData) == fieldcounter] <- action_row$RenamedTo # Rename

  } else if (WhatToDo == "Make Integer") {
    CleanedData[[fieldcounter]] <- as.integer(CleanedData[[fieldcounter]]) # Convert to integer
  }
}
# Now, pull in the Income Group Data, and join it to the raw data by CountryCode
CleanedData <- left_join(CleanedData, IncomeData, by = c("CountryCode" = "Country Code"))

# Now for initial EDA purposes, we am only going to consider the two wellbeing indexes
# which are tagged as PrimaryResponse fields.  So drop the other more granular Response fields
```

```r
# Select field names to keep
# First, all the renamed fields that weren't dropped and aren't a Response
PrimaryFields <- FieldActions %>%
  filter(Type %in% c("PrimaryResponse","Predictor","Identifier"),
         is.na(Action) | Action != "Delete") %>%
  pull(RenamedTo)
# Now add the Income Group field (since that was joined from a different data set)
PrimaryFields <- c(PrimaryFields, "Income Group")
# Subset the cleaned dataset using that list of fields
FirstCutData <- CleanedData %>%
  select(all_of(PrimaryFields))

# Done with cleaning so save the data for Preprocessing
write_xlsx(FirstCutData, path = here::here("Data","FirstCutData.xlsx"))

##############################################
#
# Exploratory Analysis starts here
#
# Summarize the data
SummaryData <- summary(FirstCutData)

# This code block breaks up the Summary data into chunks with 8 columns
# per chunk, and displays each chunk as a separate table.  This was done
# so that the Summary table can be cleanly displayed
#
cols <- ncol(SummaryData)
cols_per_table <- 8

# Loop through each block of 8
for (i in seq(1, cols, by = cols_per_table)) {
  this_section <- SummaryData[, i:min(i + cols_per_table - 1, cols)] # create sub-table

  # Print the subtable using kable to make it look nice
  cat(
    asis_output(
      kable(
        this_section,
        format = "latex",
        booktabs = TRUE,
        caption = paste0("Summary Statistics (Variables ", i, " to ",
                         min(i + cols_per_table - 1, cols), ")"),
        label = paste0("SummaryData_", i)  # <-- move label here
      ) %>%
        kable_styling(
          latex_options = c("HOLD_position", "striped", "scale_down")
        )
    )
  )
}
```

```
## \begin{table}[H]
## \centering
## \caption{\label{tab:SummaryData_1}Summary Statistics (Variables 1 to 8)}
```

```
## \centering
## \resizebox{\ifdim\width>\linewidth\linewidth\else\width\fi}{!}{
## \begin{tabular}[t]{lllllllll}
## \toprule
##   & CountryCode & CountryName &      year &   HDI\_Index &   IHDI\_Index &   GiniCoeff &   FoodIndex
## \midrule
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Length:5974} & \cellcolor{gray!10}{Length:5974} & \cellc
##  & Class :character & Class :character & 1st Qu.:2002 & 1st Qu.:0.5700 & 1st Qu.:0.3990 & 1st Qu.:11
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Mode  :character} & \cellcolor{gray!10}{Mode  :character}
##  & NA & NA & Mean   :2009 & Mean   :0.6899 & Mean   :0.5783 & Mean   :21.317 & Mean   : 91.33 & Mean
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{NA} & \cellcolor{gray!10}{NA} & \cellcolor{gray!10}{3rd
## \addlinespace
##  & NA & NA & Max.   :2023 & Max.   :0.9720 & Max.   :0.9230 & Max.   :51.891 & Max.   :578.71 & Max.
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{NA} & \cellcolor{gray!10}{NA} & \cellcolor{gray!10}{NA} &
## \bottomrule
## \end{tabular}}
## \end{table}\begin{table}[H]
## \centering
## \caption{\label{tab:SummaryData_9}Summary Statistics (Variables 9 to 16)}
## \centering
## \resizebox{\ifdim\width>\linewidth\linewidth\else\width\fi}{!}{
## \begin{tabular}[t]{lllllllll}
## \toprule
##   & ElectricAccess &   PopDensity &   PopInSlums &  WaterStress & BankingAccess &   RnDInvest & GovtE
## \midrule
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Min.   :  0.80} & \cellcolor{gray!10}{Min.   :    1.438}
##  & 1st Qu.: 63.20 & 1st Qu.:   30.875 & 1st Qu.: 1.55 & 1st Qu.:   3.7148 & 1st Qu.: 31.07 & 1st Qu.
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Median : 98.10} & \cellcolor{gray!10}{Median :   77.922}
##  & Mean   : 79.23 & Mean   :  298.200 & Mean   :28.73 & Mean   :  65.0840 & Mean   : 57.04 & Mean
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{3rd Qu.:100.00} & \cellcolor{gray!10}{3rd Qu.:  176.610}
## \addlinespace
##  & Max.   :100.00 & Max.   :18822.891 & Max.   :99.81 & Max.   :3850.5000 & Max.   :100.00 & Max.
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{NA's   :627} & \cellcolor{gray!10}{NA's   :583} & \cellc
## \bottomrule
## \end{tabular}}
## \end{table}\begin{table}[H]
## \centering
## \caption{\label{tab:SummaryData_17}Summary Statistics (Variables 17 to 24)}
## \centering
## \resizebox{\ifdim\width>\linewidth\linewidth\else\width\fi}{!}{
## \begin{tabular}[t]{lllllllll}
## \toprule
##   &  FixedIntSubs & InternetUsersPct & GDPPerCapGrowth &   GDPGrowth &      GDP & PoliticalStability
## \midrule
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Min.   : 0.0000} & \cellcolor{gray!10}{Min.   :  0.00} &
##  & 1st Qu.: 0.2673 & 1st Qu.:  2.81 & 1st Qu.:  0.07708 & 1st Qu.:  1.464 & 1st Qu.:5.577e+09 & 1st
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Median : 3.8683} & \cellcolor{gray!10}{Median : 21.00} &
##  & Mean   :10.5358 & Mean   : 32.58 & Mean   :  2.19326 & Mean   :  3.630 & Mean   :3.372e+11 & Mean
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{3rd Qu.:18.4017} & \cellcolor{gray!10}{3rd Qu.: 61.40} &
## \addlinespace
##  & Max.   :62.2147 & Max.   :100.00 & Max.   :140.49058 & Max.   :149.973 & Max.   :2.206e+13 & Max.
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{NA's   :2041} & \cellcolor{gray!10}{NA's   :675} & \cell
## \bottomrule
## \end{tabular}}
```

```
## \end{table}\begin{table}[H]
## \centering
## \caption{\label{tab:SummaryData_25}Summary Statistics (Variables 25 to 32)}
## \centering
## \resizebox{\ifdim\width>\linewidth\linewidth\else\width\fi}{!}{
## \begin{tabular}[t]{lllllllll}
## \toprule
##   & GovtEduSpendPctGDP & MortalityFromDirtyness & UHCServiceCoverage & HealthSpendPerCapita & GovtHea
## \midrule
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Min.   : 0.000} & \cellcolor{gray!10}{Min.   :  0.40} & \
##  & 1st Qu.: 3.139 & 1st Qu.:  2.95 & 1st Qu.:44.00 & 1st Qu.:   63.631 & 1st Qu.:  21.9384 & 1st Qu.
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Median : 4.232} & \cellcolor{gray!10}{Median :  6.10} & \
##  & Mean   : 4.418 & Mean   : 16.96 & Mean   :59.39 & Mean   :  959.978 & Mean   : 648.7002 & Mean
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{3rd Qu.: 5.428} & \cellcolor{gray!10}{3rd Qu.: 25.55} & \
## \addlinespace
##  & Max.   :15.863 & Max.   :108.10 & Max.   :91.00 & Max.   :12434.434 & Max.   :7871.2450 & Max.
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{NA's   :2288} & \cellcolor{gray!10}{NA's   :5791} & \cel
## \bottomrule
## \end{tabular}}
## \end{table}\begin{table}[H]
## \centering
## \caption{\label{tab:SummaryData_33}Summary Statistics (Variables 33 to 35)}
## \centering
## \resizebox{\ifdim\width>\linewidth\linewidth\else\width\fi}{!}{
## \begin{tabular}[t]{llll}
## \toprule
##   & VoiceAccountability &   Homicides & Income Group\\
## \midrule
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Min.   :-2.31340} & \cellcolor{gray!10}{Min.   :  0.000}
##  & 1st Qu.:-0.88313 & 1st Qu.:  1.228 & Class :character\\
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{Median :-0.00619} & \cellcolor{gray!10}{Median :  2.861}
##  & Mean   :-0.04756 & Mean   :  7.939 & NA\\
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{3rd Qu.: 0.85004} & \cellcolor{gray!10}{3rd Qu.:  8.879}
## \addlinespace
##  & Max.   : 1.80099 & Max.   :138.774 & NA\\
## \cellcolor{gray!10}{} & \cellcolor{gray!10}{NA's   :1112} & \cellcolor{gray!10}{NA's   :2755} & \cel
## \bottomrule
## \end{tabular}}
## \end{table}
```

```r
# Show summary statistics using skim
skim(FirstCutData)
```

Table 1: Data summary

Name	FirstCutData
Number of rows	5974
Number of columns	35
Column type frequency:	
character	3
numeric	32
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
CountryCode	0	1.00	3	9	0	206	0
CountryName	319	0.95	4	38	0	195	0
Income Group	377	0.94	10	19	0	4	0

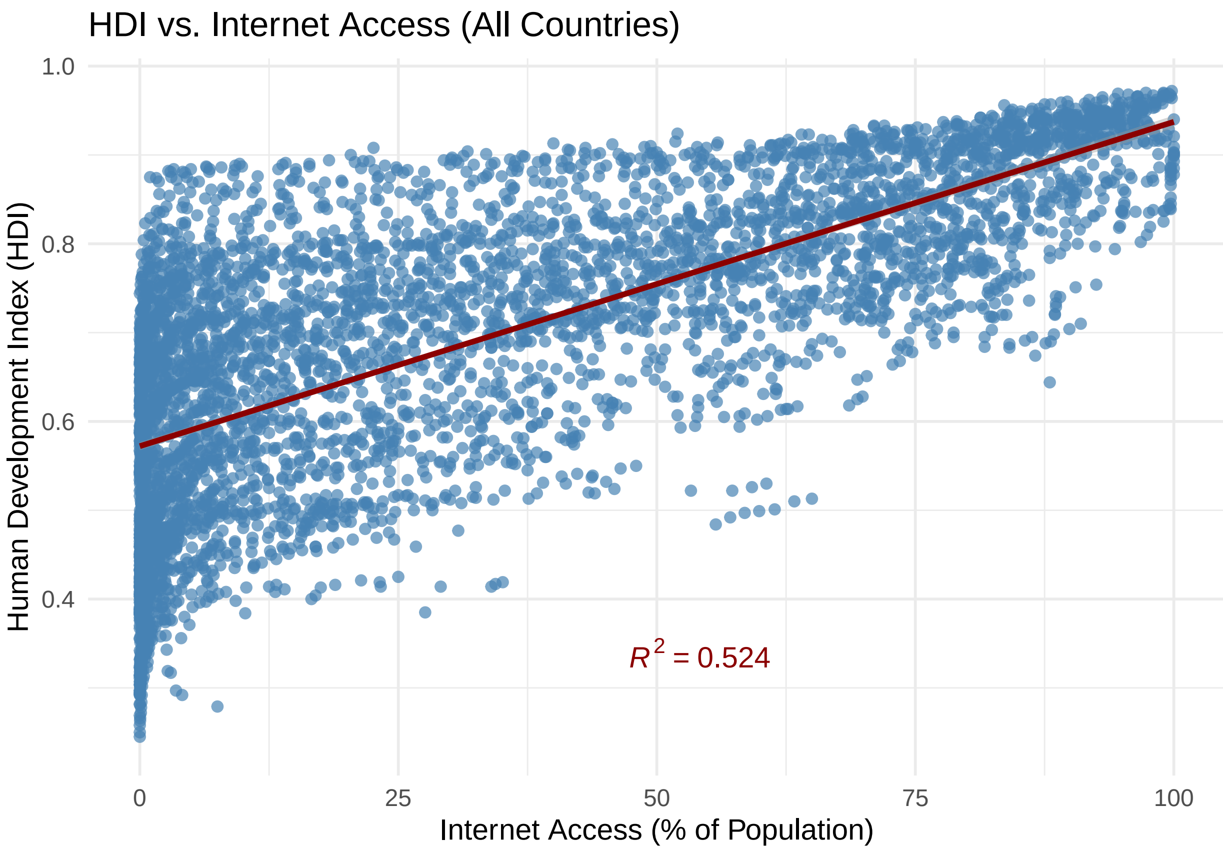
Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1.00	2.009000e+03	8.370000e+00	1995.00	2002.00	2.009000e+03	2.016000e+03	2.023000e+03	
HDI_Index	422	0.93	6.900000e-01	1.600000e-01	0.24	0.57	7.100000e-01	8.100000e-01	9.700000e-01	
IHDI_Index	3677	0.38	5.800000e-01	2.000000e-01	0.13	0.40	5.900000e-01	7.500000e-01	9.200000e-01	
GiniCoeff	3670	0.39	2.132000e+01	1.135000e+01	4.82	11.06	1.918000e+01	3.188000e+01	5.189000e+01	
FoodIndex	720	0.88	9.133000e+01	2.806000e+01	17.00	76.57	9.452000e+01	1.028000e+02	5.787100e+02	
CorruptionScore	1204	0.80	-6.000000e-02	1.000000e+00	-1.97	-0.82	-3.000000e-01	5.900000e-01	2.460000e+00	
ElectricAccess	627	0.90	7.923000e+01	3.010000e+01	0.80	63.20	9.810000e+01	1.000000e+02	1.000000e+02	
PopDensity	583	0.90	2.982000e+02	1.289800e+03	1.44	30.87	7.792000e+01	1.766100e+02	1.822289e+04	
PopInSlums	4011	0.33	2.873000e+01	2.732000e+01	0.00	1.55	2.118000e+01	5.280000e+01	9.981000e+01	
WaterStress	1505	0.75	6.508000e+02	2.666800e+03	2.03	3.71	1.118000e+01	3.970000e+01	3.850500e+03	
BankingAccess	5412	0.09	5.704000e+01	3.013000e+01	0.40	31.06	5.405000e+01	8.676000e+01	1.000000e+02	
RnDInvest	3666	0.39	9.700000e-01	9.800000e-01	0.01	0.24	6.000000e-01	1.390000e+00	6.020000e+00	
GovtEffectiveness	1223	0.80	-6.000000e-02	1.000000e+00	-2.44	-0.79	-2.000000e-01	5.900000e-01	2.470000e+00	
ShippingIndex	3592	0.40	2.514000e+01	2.522000e+01	0.60	7.54	1.517000e+01	3.416000e+01	1.711800e+02	
FixedIntSubs	2041	0.66	1.054000e+01	1.307000e+01	0.00	0.27	3.870000e+00	1.840000e+01	6.221000e+01	
InternetUsersPct	675	0.89	3.258000e+01	3.211000e+01	0.00	2.81	2.100000e+01	6.140000e+01	1.000000e+02	
GDPPerCapGrowth	443	0.93	2.190000e+00	6.020000e+00	-49.13	0.08	2.240000e+00	4.420000e+00	1.404900e+02	
GDPGrowth	443	0.93	3.630000e+00	6.160000e+00	-50.34	1.46	3.760000e+00	6.010000e+00	1.499700e+02	
GDP	471	0.92	3.371605e+11	1.471154e+12	[illegible]	[illegible]	2.460523e+10	1.584701e+11	2.106258e+13	
PoliticalStability	1160	0.81	-6.000000e-02	1.000000e+00	-3.31	-0.70	3.000000e-02	7.900000e-01	1.760000e+00	
RuleOfLaw	1114	0.81	-6.000000e-02	9.900000e-01	-2.59	-0.82	-2.100000e-01	6.900000e-01	2.120000e+00	
RqdEduYears	1432	0.76	9.420000e+00	2.250000e+00	4.00	8.00	9.000000e+00	1.000000e+01	1.700000e+01	
GovtEduSpendPctGDP	2288	0.62	4.420000e+00	1.890000e+00	0.00	3.14	4.230000e+00	5.430000e+00	1.586000e+01	
MortalityFromDistress	5791	0.03	1.696000e+01	2.165000e+01	0.40	2.95	6.100000e+00	2.555000e+01	1.081000e+02	
UHCServiceCoverage	4630	0.22	5.939000e+01	1.902000e+01	11.00	44.00	6.300000e+01	7.500000e+01	9.100000e+01	
HealthSpendPerCapita	1617	0.73	9.599800e+02	1.689180e+03	4.18	63.63	2.612900e+02	8.904800e+02	1.243443e+04	
GovtHealthSpendPerCapita	1636	0.73	6.487000e+02	1.235080e+03	3.15	21.94	1.282400e+02	5.273600e+02	7.871240e+03	
MigrantsPct	5009	0.16	8.790000e+00	1.358000e+01	0.04	1.21	3.660000e+00	1.063000e+01	8.840000e+01	
FoodInsecurityPct	4953	0.17	2.909000e+01	2.301000e+01	2.00	8.90	2.230000e+01	4.530000e+01	8.920000e+01	
RuralPopulGrowth	458	0.92	3.900000e-01	3.700000e+00	-235.93	-0.58	4.600000e-01	1.620000e+00	1.503000e+01	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
VoiceAccountability	1112	0.81	-5.000000e-02	1.000000e+00	-2.31	-0.88	-1.000000e-02	8.500000e-01	1.800000e+00	
Homicides	2755	0.54	7.940000e+00	1.247000e+01	0.00	1.23	2.860000e+00	8.880000e+00	1.387700e+02	

```r
# Display a scatterplot of Internet Access versus HDI with trendline and R-squared
ggplot(FirstCutData, aes(x = InternetUsersPct, y = HDI_Index)) +
  geom_point(alpha = 0.7, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +
  stat_poly_eq(
    aes(label = after_stat(rr.label)),
    formula = y ~ x,
    parse = TRUE,
    output.type = "expression",
    rr.digits = 3,
    label.x = 0.6,            # X-position (adjust if needed)
    label.y = 0.15,          # Y-position (bottom-ish)
    color = "darkred",       # Match regression line
    na.rm = TRUE
  ) +
  labs(
    title = "HDI vs. Internet Access (All Countries)",
    x = "Internet Access (% of Population)",
    y = "Human Development Index (HDI)"
  ) +
  theme_minimal()
```

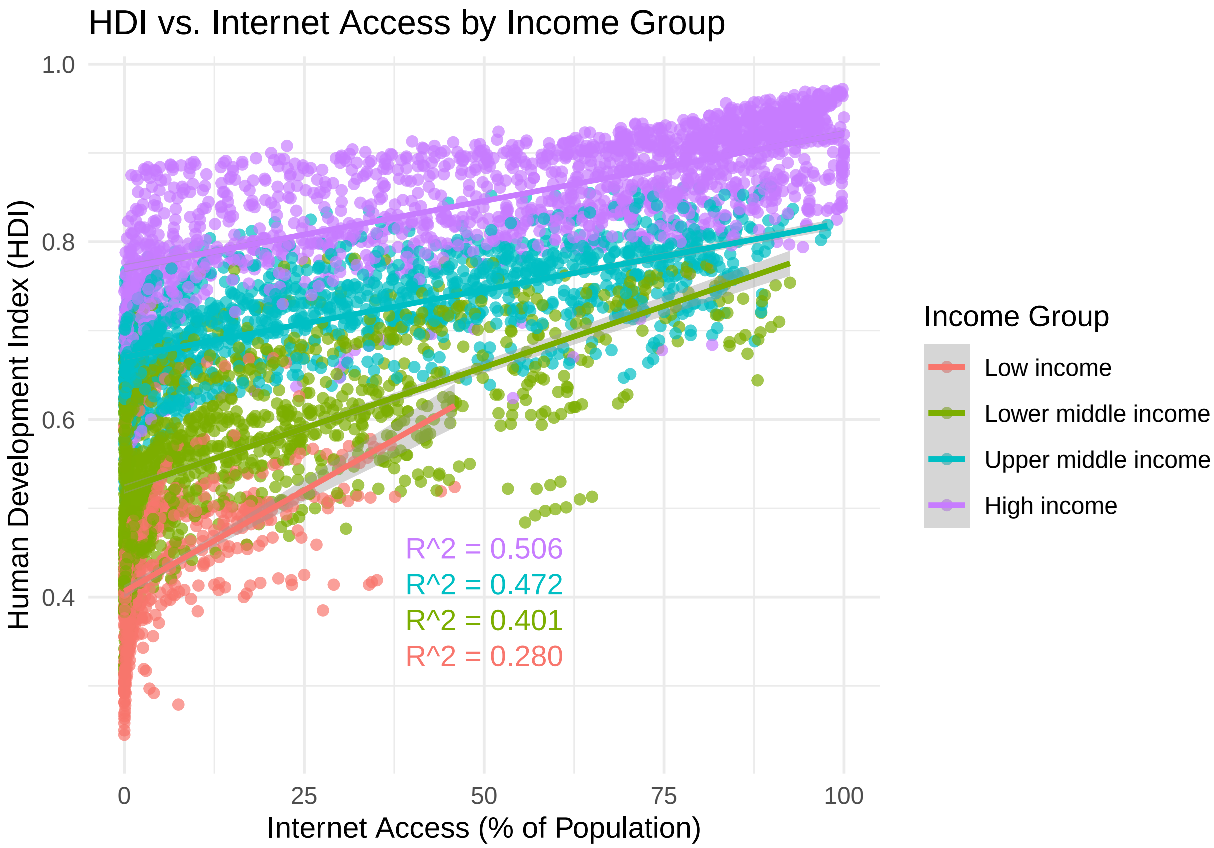
```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 978 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 978 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
# Same plot, but break out data, color differently with separate trendlines and R-squareds,
# by Income Group
FirstCutDataFiltered <- FirstCutData[!is.na(FirstCutData$`Income Group`), ]
FirstCutDataFiltered$`Income Group` <- factor(
  FirstCutDataFiltered$`Income Group`,
  levels = c("Low income", "Lower middle income", "Upper middle income", "High income")
)
ggplot(FirstCutDataFiltered, aes(x = InternetUsersPct, y = HDI_Index, color = `Income Group`)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE) +
  stat_poly_eq(
    aes(
      label = after_stat(rr.label),
      group = `Income Group`,
      color = `Income Group`
    ),
    formula = y ~ x,
    parse = FALSE,
    output.type = "text",
    rr.digits = 3,
    label.x = 0.6,
    label.y = c(0.15, 0.2, 0.25, 0.3),  # Stack in bottom right
    na.rm = TRUE
  ) +
  labs(
    title = "HDI vs. Internet Access by Income Group",
    x = "Internet Access (% of Population)",
```

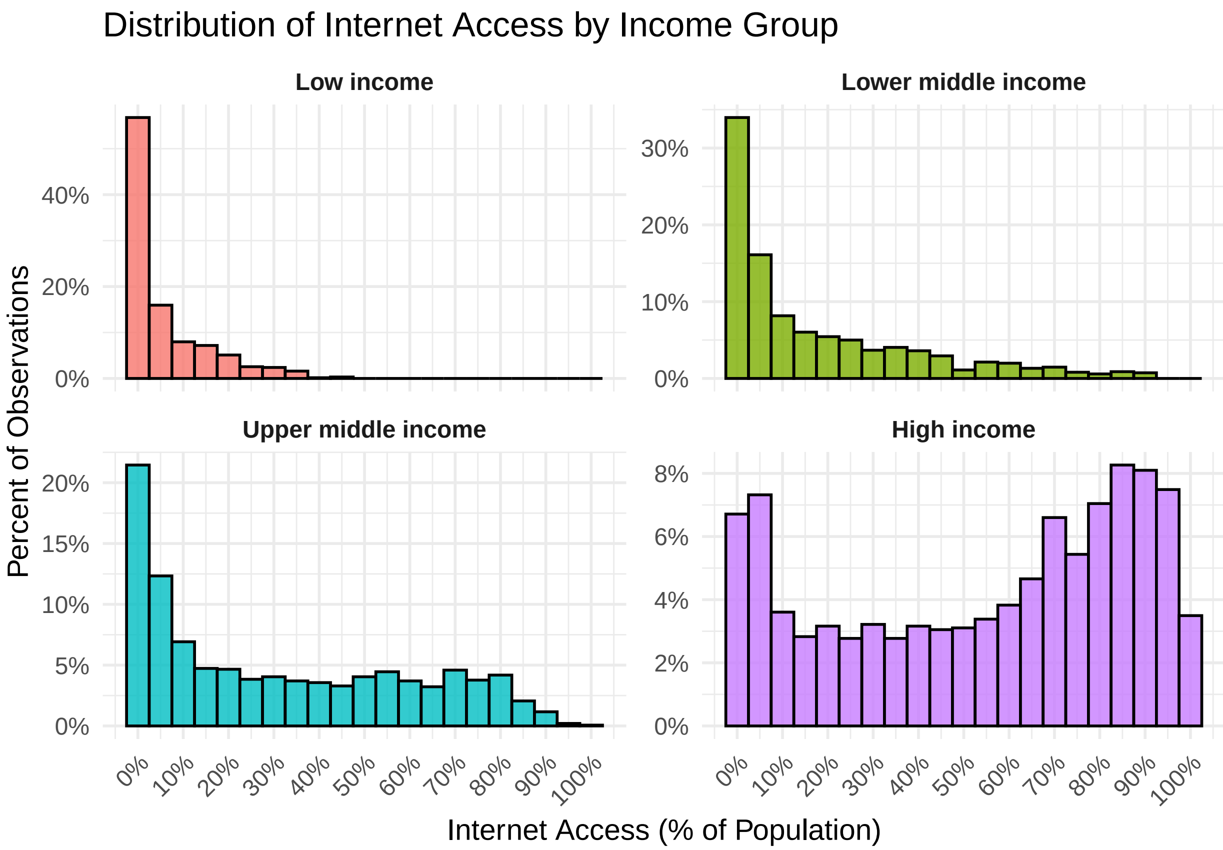
```
    y = "Human Development Index (HDI)",
    color = "Income Group"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 646 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 646 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



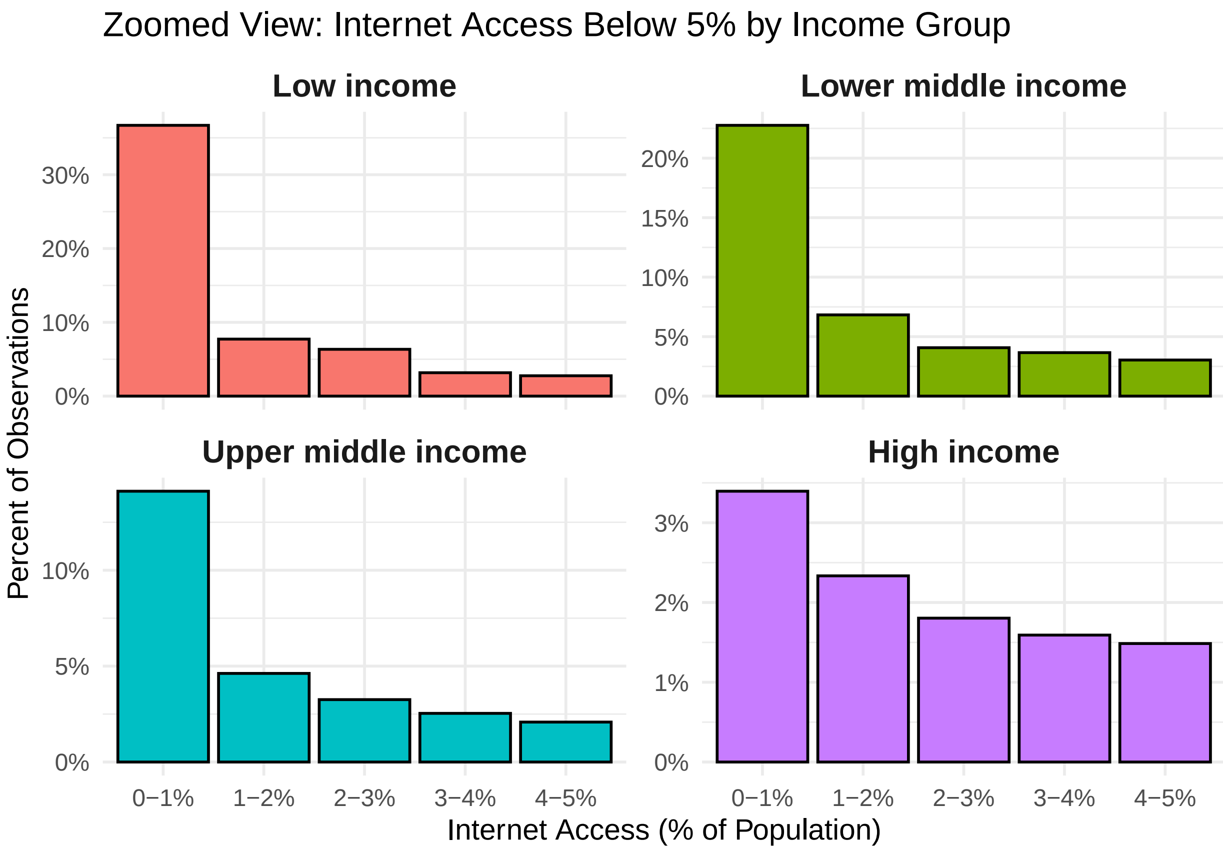

```
# Show distribution of data by Internet access in 10% increments, by Income Group
FirstCutDataFiltered <- FirstCutData[!is.na(FirstCutData$`Income Group`), ]
FirstCutDataFiltered$`Income Group` <- factor(
  FirstCutDataFiltered$`Income Group`,
  levels = c("Low income", "Lower middle income", "Upper middle income", "High income")
)

FirstCutDataFilteredClean <- FirstCutDataFiltered %>%
  filter(!is.na(InternetUsersPct))

ggplot(FirstCutDataFilteredClean, aes(x = InternetUsersPct, fill = `Income Group`)) +
  geom_histogram(
    aes(y = after_stat(count / tapply(count, PANEL, sum)[as.integer(PANEL)])),
    binwidth = 5,
    color = "black",
```

```r
    alpha = 0.8
  ) +
  facet_wrap(~ `Income Group`, scales = "free_y") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  scale_x_continuous(
    breaks = seq(0, 100, by = 10),
    labels = function(x) paste0(x, "%")
  ) +
  labs(
    title = "Distribution of Internet Access by Income Group",
    x = "Internet Access (% of Population)",
    y = "Percent of Observations"
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )
```




```r
#
#There appears to be a severe left skew problem - we should isolate how much of that is clustered
# at or near 0 to see if we need to transform it for modeling

# Step 1: Filter to only rows with Internet Access under 5%

# Bin and summarize within the 0%-5% range
```

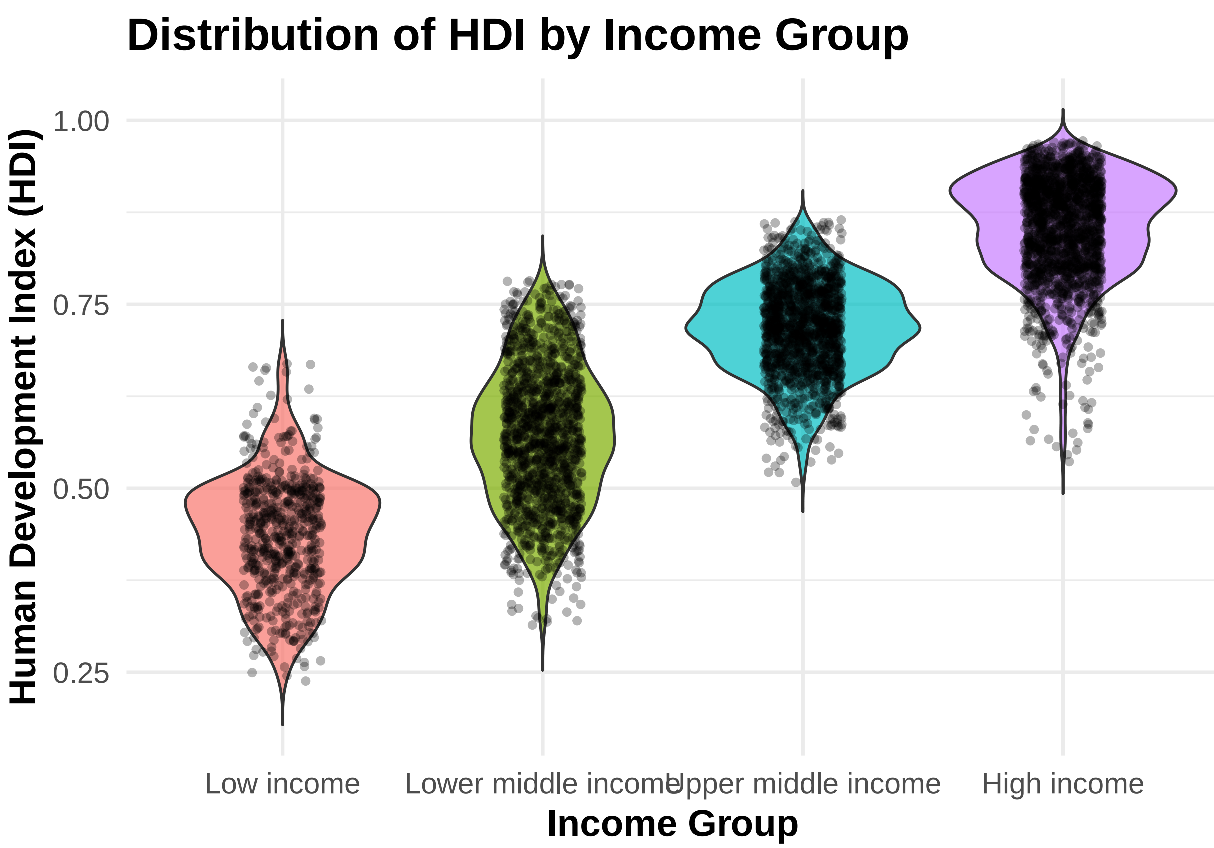
```r
zoom_data <- FirstCutDataFiltered %>%
  filter(InternetUsersPct >= 0, InternetUsersPct < 5) %>%
  mutate(Bin = cut(
    InternetUsersPct,
    breaks = seq(0, 5, by = 1),
    include.lowest = TRUE,
    right = FALSE,
    labels = c("0-1%", "1-2%", "2-3%", "3-4%", "4-5%")
  ))

# Group totals for each Income Group (full data)
group_totals <- FirstCutDataFiltered %>%
  count(`Income Group`, name = "GroupTotal")

# Bin counts and percentages
zoom_summary <- zoom_data %>%
  count(`Income Group`, Bin, name = "BinCount") %>%
  left_join(group_totals, by = "Income Group") %>%
  mutate(Percent = BinCount / GroupTotal)

# Plot
ggplot(zoom_summary, aes(x = Bin, y = Percent, fill = `Income Group`)) +
  geom_col(color = "black", width = 0.9) +
  facet_wrap(~ `Income Group`, scales = "free_y") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(
    title = "Zoomed View: Internet Access Below 5% by Income Group",
    x = "Internet Access (% of Population)",
    y = "Percent of Observations"
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(face = "bold", size = 12),
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    legend.position = "none"
  )
```




```
# Violin plot of HDI distribution by income group
ggplot(FirstCutDataFiltered, aes(x = `Income Group`, y = HDI_Index, fill = `Income Group`)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_jitter(width = 0.15, alpha = 0.3, color = "black", size = 1) +
  labs(
    title = "Distribution of HDI by Income Group",
    x = "Income Group",
    y = "Human Development Index (HDI)"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold"),
    axis.title.x = element_text(face = "bold"),
    axis.title.y = element_text(face = "bold")
  )
```

```
## Warning: Removed 417 rows containing non-finite outside the scale range
## (`stat_ydensity()`).
```

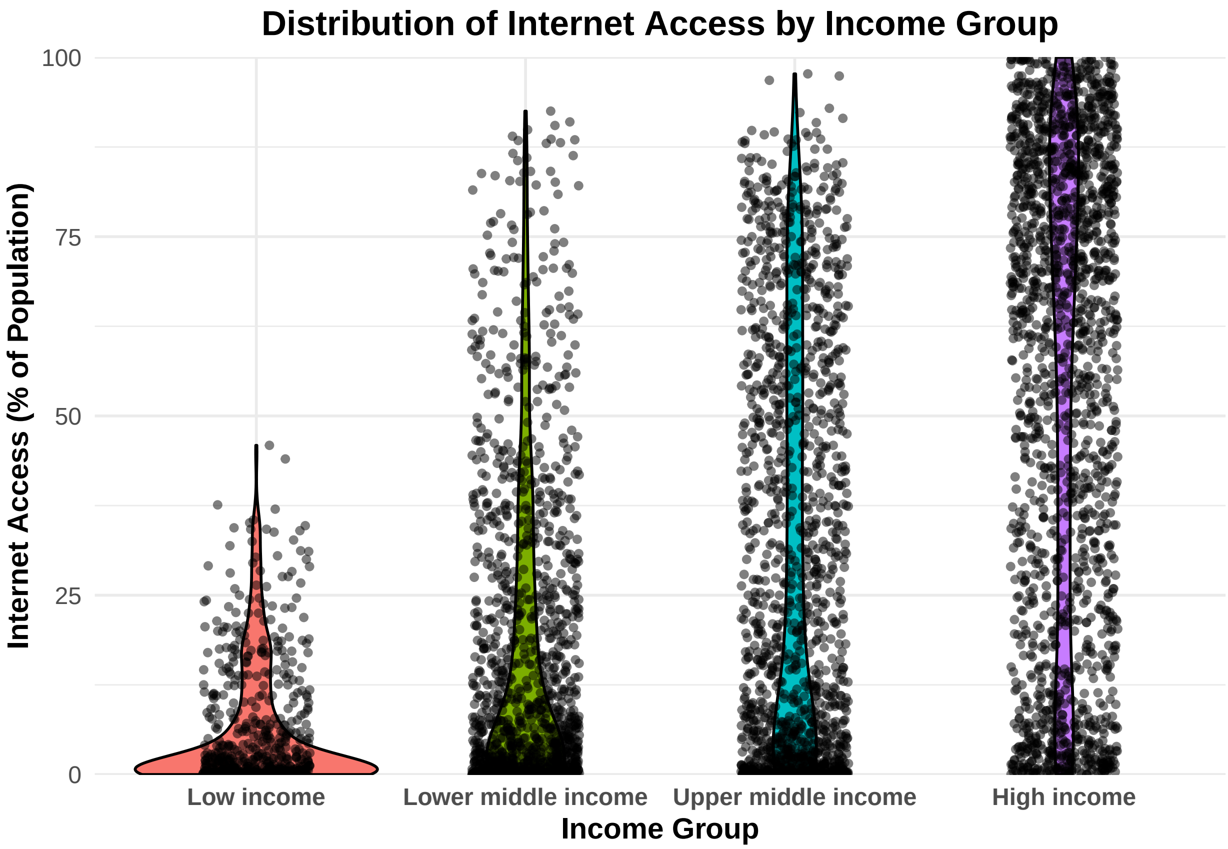
```
## Warning: Removed 417 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
Sys.sleep(2) # Wait for complicated rendering to finish

#  Violin plot of Internet access by Income Group
ggplot(FirstCutDataFiltered, aes(x = `Income Group`, y = InternetUsersPct, fill = `Income Group`)) +
  geom_violin(trim = TRUE, color = "black", adjust = 1.2) +  # trim removes tails beyond the data
  geom_jitter(width = 0.2, size = 1, alpha = 0.5, color = "black") +
  scale_y_continuous(limits = c(0, 100), expand = c(0, 0)) +  # Hard clip y-axis
  labs(
    title = "Distribution of Internet Access by Income Group",
    y = "Internet Access (% of Population)",
    x = "Income Group"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(face = "bold"),
    axis.text.x = element_text(face = "bold"),
    legend.position = "none"
  ) +
  scale_fill_manual(values = c(
    "Low income" = "#F8766D",
    "Lower middle income" = "#7CAE00",
    "Upper middle income" = "#00BFC4",
    "High income" = "#C77CFF"
  ))
```

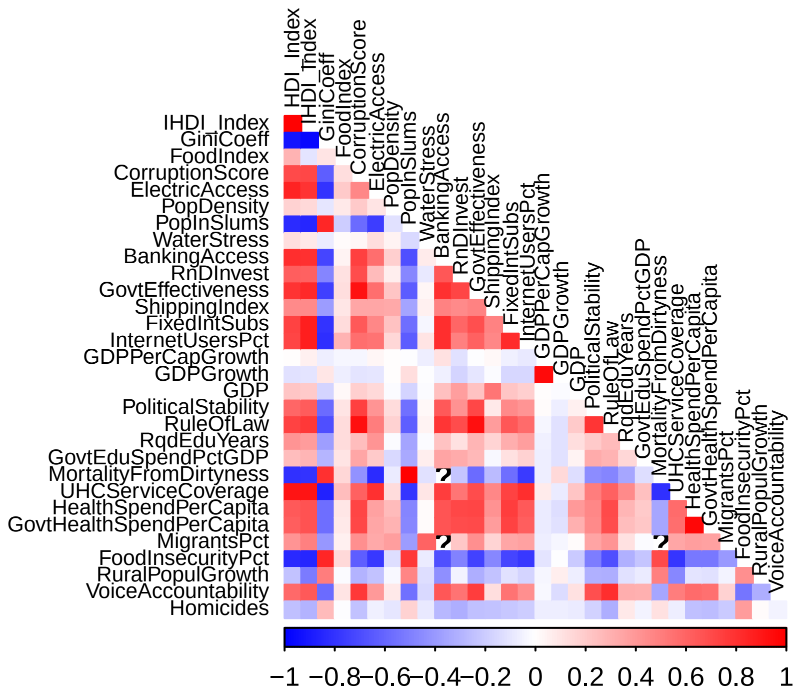
```
## Warning: Removed 348 rows containing non-finite outside the scale range
```

```
## (`stat_ydensity()`).

## Warning: Removed 370 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



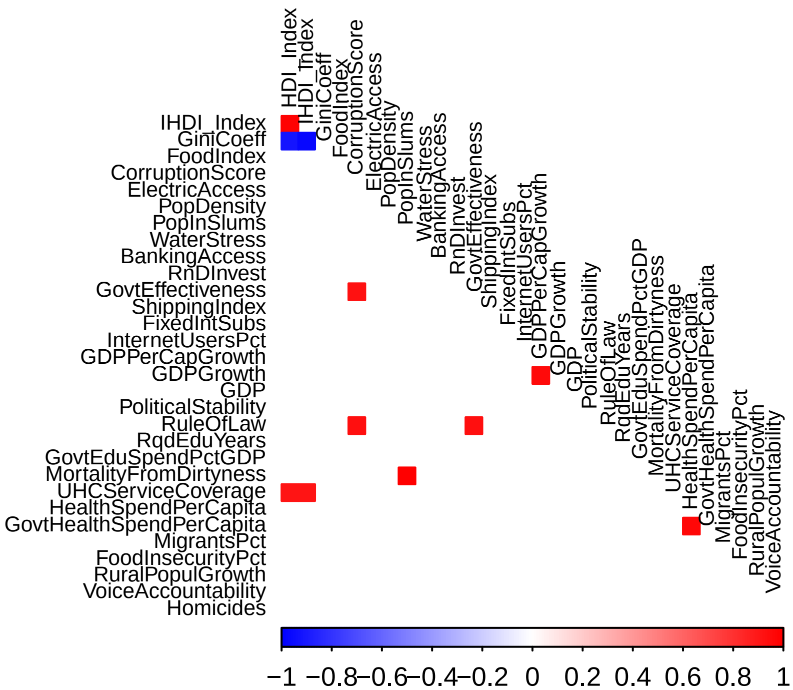

```r
Sys.sleep(2) # Wait for rendering

# Now we look at correlations to see strenght of hypothesis and find colinearity

# Step 1: Select only numeric columns and drop unwanted ones
numeric_vars <- FirstCutData %>%
  select(where(is.numeric)) %>%
  select(-year)   # Drop the 'Year' column

# Step 2: Compute correlation matrix using pairwise complete observations
cor_matrix <- cor(numeric_vars, use = "pairwise.complete.obs", method = "pearson")
cor_matrix_90 <- ifelse(abs(cor_matrix) >= 0.9, cor_matrix, NA)
cor_matrix_70_90 <- ifelse(abs(cor_matrix) >= 0.7 & abs(cor_matrix) < 0.9, cor_matrix, NA)

# Step 3: Plot correlation heatmap for ALL variables
corrplot(cor_matrix,
         method = "color",
         type = "lower",
         tl.col = "black",
         tl.cex = 0.7,
         col = colorRampPalette(c("blue", "white", "red"))(200),
         diag = FALSE)
title("Correlation Heatmap of Numeric Predictors - ALL", line = 1, cex.main = 1.2)
```

Correlation Heatmap of Numeric Predictors − ALL



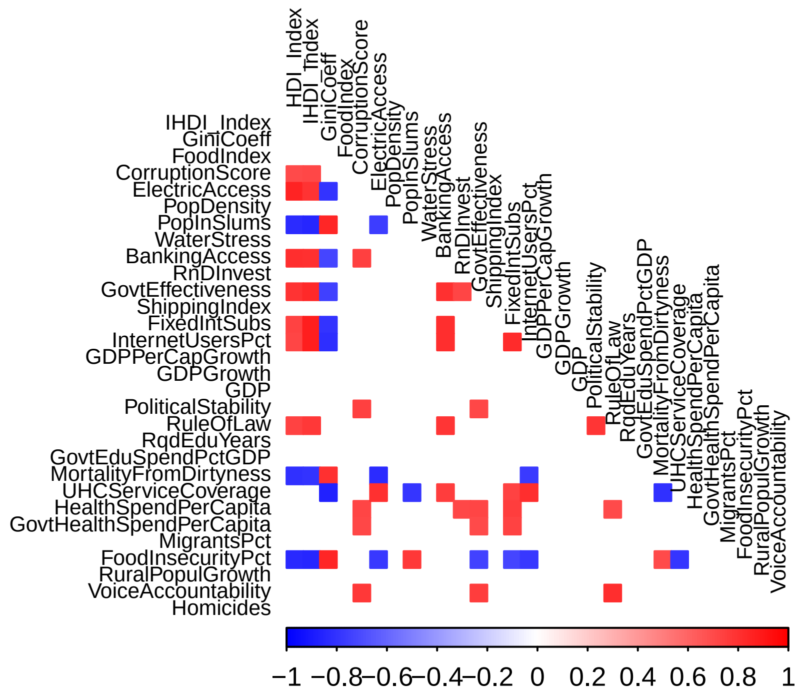
```r
# Step 4: Plot only strong correlations (90%+)
corrplot(cor_matrix_90,
         method = "color",
         type = "lower",
         tl.col = "black",
         tl.cex = 0.7,
         col = colorRampPalette(c("blue", "white", "red"))(200),
         diag = FALSE,
         na.label = " ")  # blank for missing (non-displayed) values
title("Correlation Heatmap of Numeric Predictors - 90%+ correlation (DROP)",
      line = 1, cex.main = 1.2)
```

Correlation Heatmap of Numeric Predictors − 90%+ correlation (DRO




```r
# Step 5: Plot somewhat strong correlations (70%-90%)
corrplot(cor_matrix_70_90,
         method = "color",
         type = "lower",
         tl.col = "black",
         tl.cex = 0.7,
         col = colorRampPalette(c("blue", "white", "red"))(200),
         diag = FALSE,
         na.label = " ")  # blank for missing (non-displayed) values
title("Correlation Heatmap of Numeric Predictors - 70%-90%+ correlation (VIF)",
      line = 1, cex.main = 1.2)
```

Correlation Heatmap of Numeric Predictors – 70%–90%+ correlation (




```r
# Now we will display the correlation matrix numerically

# Step 1: Compute Pearson correlation matrix using pairwise complete observations
cor_matrix <- cor(numeric_vars, use = "pairwise.complete.obs", method = "pearson")

# Step 2 Print it out, using the same "break into chunks" method and kable to make it pretty

cols <- ncol(cor_matrix)
col_names <- colnames(cor_matrix)

# Loop through and print in chunks
for (i in seq(1, cols, by = cols_per_table)) {
  # Subset the matrix
  end_col <- min(i + cols_per_table - 1, cols)
  sub_matrix <- cor_matrix[, i:end_col]

  # Print each chunk with a dynamic caption
  cat(
    asis_output(
      kable(
        round(sub_matrix, 2),
        format = "latex",
        booktabs = TRUE,
        caption = paste0("Correlation Matrix: Columns ", i, " to ", end_col)) %>%
        kable_styling(latex_options = c("HOLD_Position", "striped", "scale_down"))
    )
  )
}
```

```
## \begin{table}
```

```
## \centering
## \caption{\label{tab:Code}Correlation Matrix: Columns 1 to 8}
## \centering
## \resizebox{\ifdim\width>\linewidth\linewidth\else\width\fi}{!}{
## \begin{tabular}[t]{lrrrrrrrr}
## \toprule
##   & HDI\_Index & IHDI\_Index & GiniCoeff & FoodIndex & CorruptionScore & ElectricAccess & PopDensity
## \midrule
## \cellcolor{gray!10}{HDI\_Index} & \cellcolor{gray!10}{1.00} & \cellcolor{gray!10}{0.98} & \cellcolor
## IHDI\_Index & 0.98 & 1.00 & -0.97 & -0.10 & 0.72 & 0.80 & 0.14 & -0.85\\
## \cellcolor{gray!10}{GiniCoeff} & \cellcolor{gray!10}{-0.92} & \cellcolor{gray!10}{-0.97} & \cellcolo
## FoodIndex & 0.29 & -0.10 & 0.10 & 1.00 & 0.13 & 0.20 & 0.08 & -0.20\\
## \cellcolor{gray!10}{CorruptionScore} & \cellcolor{gray!10}{0.71} & \cellcolor{gray!10}{0.72} & \cell
## \addlinespace
## ElectricAccess & 0.85 & 0.80 & -0.79 & 0.20 & 0.47 & 1.00 & 0.10 & -0.75\\
## \cellcolor{gray!10}{PopDensity} & \cellcolor{gray!10}{0.17} & \cellcolor{gray!10}{0.14} & \cellcolor
## PopInSlums & -0.82 & -0.85 & 0.84 & -0.20 & -0.57 & -0.75 & -0.11 & 1.00\\
## \cellcolor{gray!10}{WaterStress} & \cellcolor{gray!10}{0.12} & \cellcolor{gray!10}{0.08} & \cellcolo
## BankingAccess & 0.82 & 0.81 & -0.72 & 0.04 & 0.75 & 0.56 & 0.18 & -0.69\\
## \addlinespace
## \cellcolor{gray!10}{RnDInvest} & \cellcolor{gray!10}{0.63} & \cellcolor{gray!10}{0.62} & \cellcolor{
## GovtEffectiveness & 0.79 & 0.83 & -0.74 & 0.12 & 0.92 & 0.57 & 0.22 & -0.64\\
## \cellcolor{gray!10}{ShippingIndex} & \cellcolor{gray!10}{0.46} & \cellcolor{gray!10}{0.46} & \cellco
## FixedIntSubs & 0.73 & 0.88 & -0.80 & 0.14 & 0.64 & 0.47 & 0.24 & -0.57\\
## \cellcolor{gray!10}{InternetUsersPct} & \cellcolor{gray!10}{0.72} & \cellcolor{gray!10}{0.87} & \cel
## \addlinespace
## GDPPerCapGrowth & 0.00 & 0.05 & -0.06 & -0.03 & -0.03 & 0.03 & 0.01 & 0.00\\
## \cellcolor{gray!10}{GDPGrowth} & \cellcolor{gray!10}{-0.12} & \cellcolor{gray!10}{-0.10} & \cellcolo
## GDP & 0.22 & 0.21 & -0.17 & 0.03 & 0.19 & 0.14 & -0.02 & -0.17\\
## \cellcolor{gray!10}{PoliticalStability} & \cellcolor{gray!10}{0.59} & \cellcolor{gray!10}{0.63} & \c
## RuleOfLaw & 0.74 & 0.77 & -0.68 & 0.13 & 0.94 & 0.50 & 0.18 & -0.63\\
## \addlinespace
## \cellcolor{gray!10}{RqdEduYears} & \cellcolor{gray!10}{0.41} & \cellcolor{gray!10}{0.39} & \cellcolo
## GovtEduSpendPctGDP & 0.26 & 0.28 & -0.28 & 0.11 & 0.35 & 0.20 & -0.15 & -0.33\\
## \cellcolor{gray!10}{MortalityFromDirtyness} & \cellcolor{gray!10}{-0.80} & \cellcolor{gray!10}{-0.79
## UHCServiceCoverage & 0.92 & 0.92 & -0.86 & 0.25 & 0.61 & 0.81 & 0.13 & -0.79\\
## \cellcolor{gray!10}{HealthSpendPerCapita} & \cellcolor{gray!10}{0.64} & \cellcolor{gray!10}{0.68} &
## \addlinespace
## GovtHealthSpendPerCapita & 0.63 & 0.67 & -0.57 & 0.09 & 0.72 & 0.35 & 0.29 & -0.47\\
## \cellcolor{gray!10}{MigrantsPct} & \cellcolor{gray!10}{0.41} & \cellcolor{gray!10}{0.50} & \cellcolo
## FoodInsecurityPct & -0.83 & -0.85 & 0.84 & 0.14 & -0.62 & -0.78 & -0.13 & 0.77\\
## \cellcolor{gray!10}{RuralPopulGrowth} & \cellcolor{gray!10}{-0.24} & \cellcolor{gray!10}{-0.52} & \c
## VoiceAccountability & 0.59 & 0.65 & -0.58 & 0.10 & 0.77 & 0.40 & 0.08 & -0.55\\
## \addlinespace
## \cellcolor{gray!10}{Homicides} & \cellcolor{gray!10}{-0.24} & \cellcolor{gray!10}{-0.30} & \cellcolo
## \bottomrule
## \end{tabular}}
## \end{table}\begin{table}
## \centering
## \caption{\label{tab:Code}Correlation Matrix: Columns 9 to 16}
## \centering
## \resizebox{\ifdim\width>\linewidth\linewidth\else\width\fi}{!}{
## \begin{tabular}[t]{lrrrrrrrr}
## \toprule
```

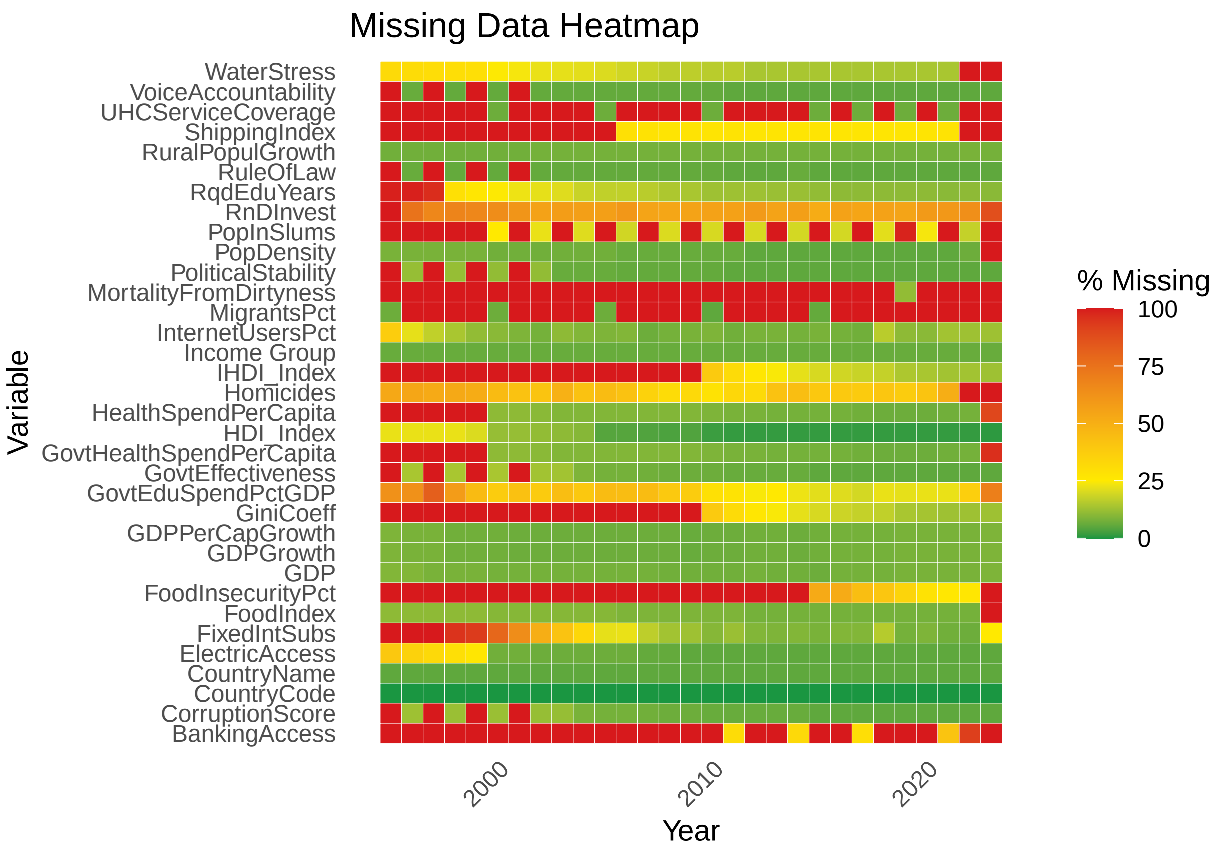
```
##   & WaterStress & BankingAccess & RnDInvest & GovtEffectiveness & ShippingIndex & FixedIntSubs & Inte
## \midrule
## \cellcolor{gray!10}{HDI\_Index} & \cellcolor{gray!10}{0.12} & \cellcolor{gray!10}{0.82} & \cellcolor
## IHDI\_Index & 0.08 & 0.81 & 0.62 & 0.83 & 0.46 & 0.88 & 0.87 & 0.05\\
## \cellcolor{gray!10}{GiniCoeff} & \cellcolor{gray!10}{-0.08} & \cellcolor{gray!10}{-0.72} & \cellcolo
## FoodIndex & 0.01 & 0.04 & 0.12 & 0.12 & 0.10 & 0.14 & 0.29 & -0.03\\
## \cellcolor{gray!10}{CorruptionScore} & \cellcolor{gray!10}{0.04} & \cellcolor{gray!10}{0.75} & \cell
## \addlinespace
## ElectricAccess & 0.13 & 0.56 & 0.27 & 0.57 & 0.34 & 0.47 & 0.55 & 0.03\\
## \cellcolor{gray!10}{PopDensity} & \cellcolor{gray!10}{0.05} & \cellcolor{gray!10}{0.18} & \cellcolor
## PopInSlums & -0.14 & -0.69 & -0.47 & -0.64 & -0.36 & -0.57 & -0.64 & 0.00\\
## \cellcolor{gray!10}{WaterStress} & \cellcolor{gray!10}{1.00} & \cellcolor{gray!10}{0.08} & \cellcolo
## BankingAccess & 0.08 & 1.00 & 0.66 & 0.80 & 0.49 & 0.81 & 0.81 & 0.12\\
## \addlinespace
## \cellcolor{gray!10}{RnDInvest} & \cellcolor{gray!10}{-0.08} & \cellcolor{gray!10}{0.66} & \cellcolor
## GovtEffectiveness & 0.04 & 0.80 & 0.72 & 1.00 & 0.49 & 0.69 & 0.62 & -0.01\\
## \cellcolor{gray!10}{ShippingIndex} & \cellcolor{gray!10}{0.05} & \cellcolor{gray!10}{0.49} & \cellco
## FixedIntSubs & -0.04 & 0.81 & 0.59 & 0.69 & 0.49 & 1.00 & 0.83 & -0.06\\
## \cellcolor{gray!10}{InternetUsersPct} & \cellcolor{gray!10}{0.10} & \cellcolor{gray!10}{0.81} & \cel
## \addlinespace
## GDPPerCapGrowth & -0.07 & 0.12 & -0.12 & -0.01 & 0.03 & -0.06 & -0.08 & 1.00\\
## \cellcolor{gray!10}{GDPGrowth} & \cellcolor{gray!10}{-0.01} & \cellcolor{gray!10}{-0.05} & \cellcolo
## GDP & -0.01 & 0.24 & 0.37 & 0.23 & 0.53 & 0.22 & 0.19 & 0.01\\
## \cellcolor{gray!10}{PoliticalStability} & \cellcolor{gray!10}{0.03} & \cellcolor{gray!10}{0.65} & \c
## RuleOfLaw & 0.05 & 0.78 & 0.70 & 0.94 & 0.39 & 0.65 & 0.57 & -0.03\\
## \addlinespace
## \cellcolor{gray!10}{RqdEduYears} & \cellcolor{gray!10}{-0.02} & \cellcolor{gray!10}{0.23} & \cellcol
## GovtEduSpendPctGDP & 0.06 & 0.36 & 0.32 & 0.27 & -0.02 & 0.12 & 0.17 & -0.07\\
## \cellcolor{gray!10}{MortalityFromDirtyness} & \cellcolor{gray!10}{-0.12} & \cellcolor{gray!10}{NA} &
## UHCServiceCoverage & 0.10 & 0.76 & 0.54 & 0.70 & 0.46 & 0.73 & 0.82 & 0.07\\
## \cellcolor{gray!10}{HealthSpendPerCapita} & \cellcolor{gray!10}{0.01} & \cellcolor{gray!10}{0.66} &
## \addlinespace
## GovtHealthSpendPerCapita & 0.02 & 0.66 & 0.69 & 0.71 & 0.39 & 0.74 & 0.63 & -0.07\\
## \cellcolor{gray!10}{MigrantsPct} & \cellcolor{gray!10}{0.61} & \cellcolor{gray!10}{NA} & \cellcolor{
## FoodInsecurityPct & -0.08 & -0.69 & -0.49 & -0.74 & -0.47 & -0.72 & -0.77 & -0.11\\
## \cellcolor{gray!10}{RuralPopulGrowth} & \cellcolor{gray!10}{-0.13} & \cellcolor{gray!10}{-0.44} & \c
## VoiceAccountability & -0.14 & 0.65 & 0.54 & 0.75 & 0.13 & 0.54 & 0.44 & -0.02\\
## \addlinespace
## \cellcolor{gray!10}{Homicides} & \cellcolor{gray!10}{-0.09} & \cellcolor{gray!10}{-0.28} & \cellcolo
## \bottomrule
## \end{tabular}}
## \end{table}\begin{table}
## \centering
## \caption{\label{tab:Code}Correlation Matrix: Columns 17 to 24}
## \centering
## \resizebox{\ifdim\width>\linewidth\linewidth\else\width\fi}{!}{
## \begin{tabular}[t]{lrrrrrrrr}
## \toprule
##   & GDPGrowth & GDP & PoliticalStability & RuleOfLaw & RqdEduYears & GovtEduSpendPctGDP & MortalityF
## \midrule
## \cellcolor{gray!10}{HDI\_Index} & \cellcolor{gray!10}{-0.12} & \cellcolor{gray!10}{0.22} & \cellcolo
## IHDI\_Index & -0.10 & 0.21 & 0.63 & 0.77 & 0.39 & 0.28 & -0.79 & 0.92\\
## \cellcolor{gray!10}{GiniCoeff} & \cellcolor{gray!10}{0.10} & \cellcolor{gray!10}{-0.17} & \cellcolor
## FoodIndex & -0.07 & 0.03 & 0.10 & 0.13 & 0.13 & 0.11 & 0.15 & 0.25\\
```

```
## \cellcolor{gray!10}{CorruptionScore} & \cellcolor{gray!10}{-0.10} & \cellcolor{gray!10}{0.19} & \cel
## \addlinespace
## ElectricAccess & -0.09 & 0.14 & 0.41 & 0.50 & 0.42 & 0.20 & -0.82 & 0.81\\
## \cellcolor{gray!10}{PopDensity} & \cellcolor{gray!10}{0.00} & \cellcolor{gray!10}{-0.02} & \cellcolo
## PopInSlums & 0.12 & -0.17 & -0.56 & -0.63 & -0.36 & -0.33 & 1.00 & -0.79\\
## \cellcolor{gray!10}{WaterStress} & \cellcolor{gray!10}{-0.01} & \cellcolor{gray!10}{-0.01} & \cellco
## BankingAccess & -0.05 & 0.24 & 0.65 & 0.78 & 0.23 & 0.36 & NA & 0.76\\
## \addlinespace
## \cellcolor{gray!10}{RnDInvest} & \cellcolor{gray!10}{-0.16} & \cellcolor{gray!10}{0.37} & \cellcolor
## GovtEffectiveness & -0.09 & 0.23 & 0.71 & 0.94 & 0.28 & 0.27 & -0.57 & 0.70\\
## \cellcolor{gray!10}{ShippingIndex} & \cellcolor{gray!10}{-0.01} & \cellcolor{gray!10}{0.53} & \cellc
## FixedIntSubs & -0.16 & 0.22 & 0.46 & 0.65 & 0.32 & 0.12 & -0.56 & 0.73\\
## \cellcolor{gray!10}{InternetUsersPct} & \cellcolor{gray!10}{-0.16} & \cellcolor{gray!10}{0.19} & \cel
## \addlinespace
## GDPPerCapGrowth & 0.96 & 0.01 & 0.01 & -0.03 & -0.05 & -0.07 & -0.06 & 0.07\\
## \cellcolor{gray!10}{GDPGrowth} & \cellcolor{gray!10}{1.00} & \cellcolor{gray!10}{-0.02} & \cellcolor
## GDP & -0.02 & 1.00 & 0.05 & 0.20 & 0.13 & 0.05 & -0.12 & 0.22\\
## \cellcolor{gray!10}{PoliticalStability} & \cellcolor{gray!10}{-0.06} & \cellcolor{gray!10}{0.05} & \
## RuleOfLaw & -0.12 & 0.20 & 0.78 & 1.00 & 0.26 & 0.29 & -0.47 & 0.62\\
## \addlinespace
## \cellcolor{gray!10}{RqdEduYears} & \cellcolor{gray!10}{-0.11} & \cellcolor{gray!10}{0.13} & \cellcol
## GovtEduSpendPctGDP & -0.11 & 0.05 & 0.33 & 0.29 & 0.16 & 1.00 & -0.13 & 0.25\\
## \cellcolor{gray!10}{MortalityFromDirtyness} & \cellcolor{gray!10}{0.14} & \cellcolor{gray!10}{-0.12}
## UHCServiceCoverage & -0.08 & 0.22 & 0.51 & 0.62 & 0.45 & 0.25 & -0.80 & 1.00\\
## \cellcolor{gray!10}{HealthSpendPerCapita} & \cellcolor{gray!10}{-0.12} & \cellcolor{gray!10}{0.40} &
## \addlinespace
## GovtHealthSpendPerCapita & -0.12 & 0.31 & 0.46 & 0.70 & 0.26 & 0.21 & -0.34 & 0.56\\
## \cellcolor{gray!10}{MigrantsPct} & \cellcolor{gray!10}{-0.03} & \cellcolor{gray!10}{0.01} & \cellcol
## FoodInsecurityPct & 0.00 & -0.20 & -0.51 & -0.68 & -0.30 & -0.22 & 0.70 & -0.79\\
## \cellcolor{gray!10}{RuralPopulGrowth} & \cellcolor{gray!10}{0.07} & \cellcolor{gray!10}{-0.07} & \cel
## VoiceAccountability & -0.13 & 0.11 & 0.68 & 0.82 & 0.32 & 0.31 & -0.34 & 0.50\\
## \addlinespace
## \cellcolor{gray!10}{Homicides} & \cellcolor{gray!10}{-0.07} & \cellcolor{gray!10}{-0.08} & \cellcolo
## \bottomrule
## \end{tabular}}
## \end{table}\begin{table}
## \centering
## \caption{\label{tab:Code}Correlation Matrix: Columns 25 to 31}
## \centering
## \resizebox{\ifdim\width>\linewidth\linewidth\else\width\fi}{!}{
## \begin{tabular}[t]{lrrrrrrr}
## \toprule
##   & HealthSpendPerCapita & GovtHealthSpendPerCapita & MigrantsPct & FoodInsecurityPct & RuralPopulGro
## \midrule
## \cellcolor{gray!10}{HDI\_Index} & \cellcolor{gray!10}{0.64} & \cellcolor{gray!10}{0.63} & \cellcolor
## IHDI\_Index & 0.68 & 0.67 & 0.50 & -0.85 & -0.52 & 0.65 & -0.30\\
## \cellcolor{gray!10}{GiniCoeff} & \cellcolor{gray!10}{-0.57} & \cellcolor{gray!10}{-0.57} & \cellcolo
## FoodIndex & 0.09 & 0.09 & 0.06 & 0.14 & -0.02 & 0.10 & -0.01\\
## \cellcolor{gray!10}{CorruptionScore} & \cellcolor{gray!10}{0.73} & \cellcolor{gray!10}{0.72} & \cell
## \addlinespace
## ElectricAccess & 0.36 & 0.35 & 0.34 & -0.78 & -0.23 & 0.40 & -0.06\\
## \cellcolor{gray!10}{PopDensity} & \cellcolor{gray!10}{0.26} & \cellcolor{gray!10}{0.29} & \cellcolor
## PopInSlums & -0.48 & -0.47 & -0.38 & 0.77 & 0.48 & -0.55 & 0.17\\
## \cellcolor{gray!10}{WaterStress} & \cellcolor{gray!10}{0.01} & \cellcolor{gray!10}{0.02} & \cellcolo
```

```
## BankingAccess & 0.66 & 0.66 & NA & -0.69 & -0.44 & 0.65 & -0.28\\
## \addlinespace
## \cellcolor{gray!10}{RnDInvest} & \cellcolor{gray!10}{0.71} & \cellcolor{gray!10}{0.69} & \cellcolor{g
## GovtEffectiveness & 0.72 & 0.71 & 0.43 & -0.74 & -0.31 & 0.75 & -0.24\\
## \cellcolor{gray!10}{ShippingIndex} & \cellcolor{gray!10}{0.43} & \cellcolor{gray!10}{0.39} & \cellco
## FixedIntSubs & 0.75 & 0.74 & 0.35 & -0.72 & -0.12 & 0.54 & -0.22\\
## \cellcolor{gray!10}{InternetUsersPct} & \cellcolor{gray!10}{0.64} & \cellcolor{gray!10}{0.63} & \cel
## \addlinespace
## GDPPerCapGrowth & -0.07 & -0.07 & -0.06 & -0.11 & -0.03 & -0.02 & -0.06\\
## \cellcolor{gray!10}{GDPGrowth} & \cellcolor{gray!10}{-0.12} & \cellcolor{gray!10}{-0.12} & \cellcolo
## GDP & 0.40 & 0.31 & 0.01 & -0.20 & -0.07 & 0.11 & -0.08\\
## \cellcolor{gray!10}{PoliticalStability} & \cellcolor{gray!10}{0.46} & \cellcolor{gray!10}{0.46} & \c
## RuleOfLaw & 0.71 & 0.70 & 0.42 & -0.68 & -0.31 & 0.82 & -0.31\\
## \addlinespace
## \cellcolor{gray!10}{RqdEduYears} & \cellcolor{gray!10}{0.28} & \cellcolor{gray!10}{0.26} & \cellcolo
## GovtEduSpendPctGDP & 0.20 & 0.21 & -0.01 & -0.22 & -0.08 & 0.31 & -0.04\\
## \cellcolor{gray!10}{MortalityFromDirtyness} & \cellcolor{gray!10}{-0.35} & \cellcolor{gray!10}{-0.34
## UHCServiceCoverage & 0.58 & 0.56 & 0.35 & -0.79 & -0.46 & 0.50 & -0.07\\
## \cellcolor{gray!10}{HealthSpendPerCapita} & \cellcolor{gray!10}{1.00} & \cellcolor{gray!10}{0.97} &
## \addlinespace
## GovtHealthSpendPerCapita & 0.97 & 1.00 & 0.37 & -0.53 & -0.11 & 0.56 & -0.26\\
## \cellcolor{gray!10}{MigrantsPct} & \cellcolor{gray!10}{0.38} & \cellcolor{gray!10}{0.37} & \cellcolo
## FoodInsecurityPct & -0.52 & -0.53 & -0.39 & 1.00 & 0.43 & -0.54 & 0.39\\
## \cellcolor{gray!10}{RuralPopulGrowth} & \cellcolor{gray!10}{-0.10} & \cellcolor{gray!10}{-0.11} & \c
## VoiceAccountability & 0.57 & 0.56 & 0.17 & -0.54 & -0.31 & 1.00 & -0.04\\
## \addlinespace
## \cellcolor{gray!10}{Homicides} & \cellcolor{gray!10}{-0.24} & \cellcolor{gray!10}{-0.26} & \cellcolo
## \bottomrule
## \end{tabular}}
## \end{table}
```

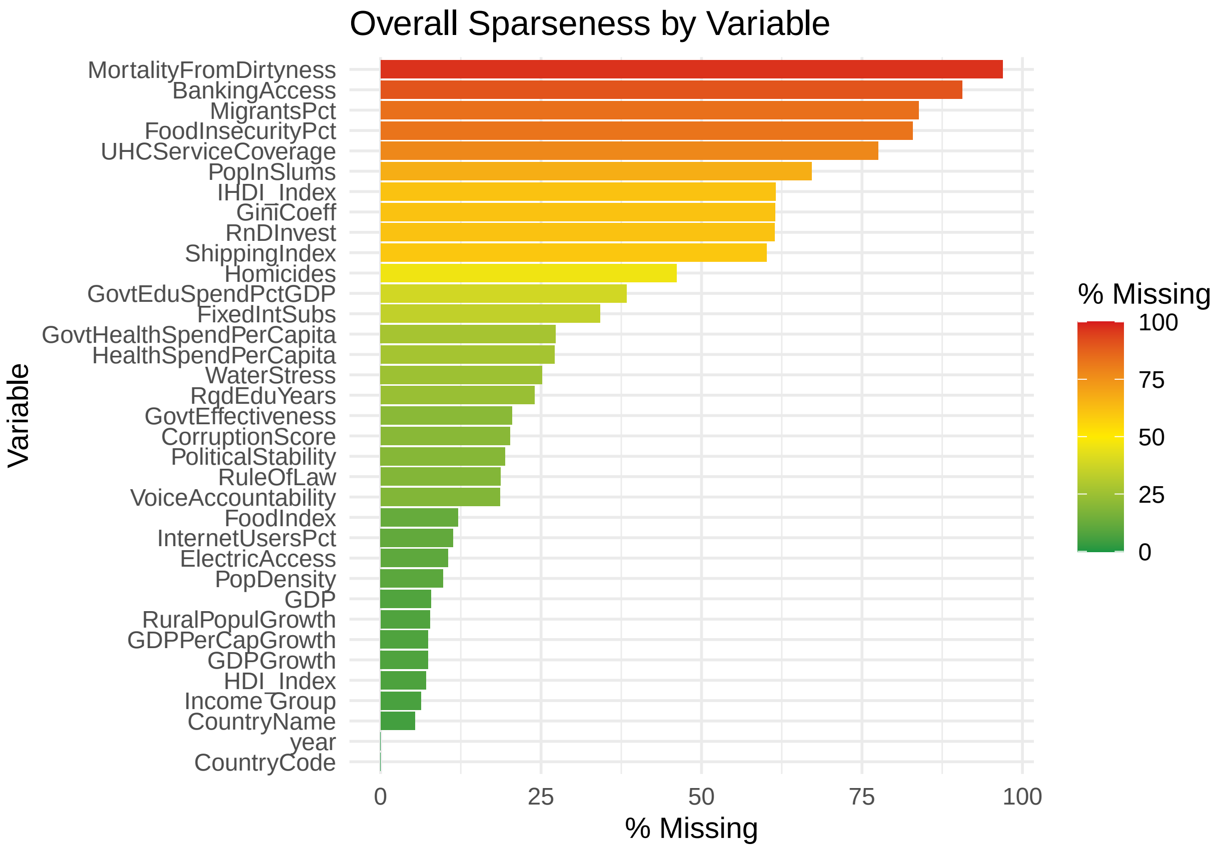
```r
#
# Now let's look at missingness by creating a "stoplight color0-schemed" heatmap for all
# variables by year as a starting point for our imputation and dropping efforts

# Step 1: Compute % missing per variable per year
missing_heatmap_data <- FirstCutData %>%
  group_by(year) %>%
  summarize(across(everything(), ~mean(is.na(.)) * 100)) %>%
  pivot_longer(-year, names_to = "variable", values_to = "pct_missing")

# Step 2: Plot with stoplight colors
stoplight <- c("#1a9641", "#ffea00", "#d7191c")
print(ggplot(missing_heatmap_data, aes(x = year, y = variable, fill = pct_missing)) +
        geom_tile(color = "white") +
        scale_fill_gradientn(
          colors = stoplight,
          values = scales::rescale(c(0, 25, 100)),
          limits = c(0, 100),
          name = "% Missing"
        ) +
        labs(
          title = "Missing Data Heatmap",
          x = "Year",
```

```
    y = "Variable"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid = element_blank()
  ))
```




```
# Now show a cut by variable
FirstCutData %>%
  summarize(across(everything(), ~ mean(is.na(.)) * 100)) %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "MissingPct") %>%
  ggplot(aes(x = reorder(Variable, MissingPct), y = MissingPct, fill = MissingPct)) +
  geom_col() +
  coord_flip() +
  scale_fill_gradientn(
    colors = stoplight,
    limits = c(0, 100),
    breaks = c(0, 25, 50, 75, 100),  # include 0 and 100
    labels = c("0", "25", "50", "75", "100"),
    name = "% Missing"
  ) +
  labs(
    title = "Overall Sparseness by Variable",
    x = "Variable",
    y = "% Missing"
  ) +
```

```r
  theme_minimal()
```




```r
# Now show a cut by Year
FirstCutData %>%
  group_by(year) %>%
  summarize(across(everything(), ~ mean(is.na(.)) * 100)) %>%
  pivot_longer(-year, names_to = "variable", values_to = "pct_missing") %>%
  group_by(year) %>%
  summarize(pct_missing = mean(pct_missing)) %>%
  arrange(year) %>%
  ggplot(aes(y = factor(year), x = pct_missing, fill = pct_missing)) +
  geom_col() +
  scale_fill_gradientn(
    colors = stoplight,
    values = rescale(c(0, 50, 100)),
    name = "% Missing",
    limits = c(0, 100),
    breaks = c(0, 50, 100),
    labels = c("0", "50", "100")
  ) +
  labs(
    y = "Year",
    x = "% Missing (All Variables)",
    title = "Data Sparseness by Year"
  ) +
  theme_minimal() +
  theme(
```

```
  axis.text.y = element_text(size = 8),
  legend.position = "right"
)
```